Maryland Spending Analysis

Fiscal Year: 2008-2026

394,817 Vendor Payments

$56.175B in Total Spending

Maryland State House, Annapolis Maryland

Prepared by Ryan Eller

PROJECT OVERVIEW

The Challenge

The Problem

Raw government spending data is hard to interpret — unclear where payments go, how spending changes, or in-state vs. out-of-state distribution.

The Solution

Cleaned & enriched dataset in Python, added ZIP-based location data, built interactive Tableau dashboard.

DATASET

Data at a Glance

394K

Payment Records

145

Agencies

60K+

Vendors

18

Final Columns

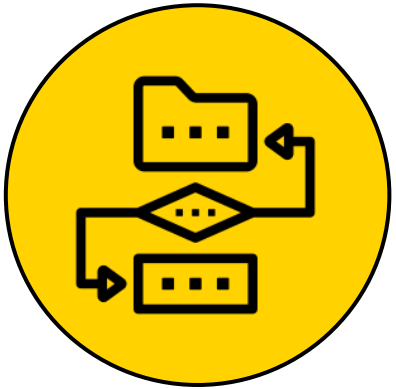
Up from **8** after my data enrichment

Source: Open Data Portal: State of Maryland Payments Data

DATA CLEANING

Python Data Preparation

Standardize Names



Convert Data Types

Clean Zip Codes

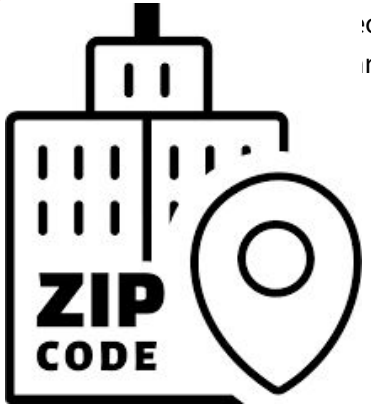
Data Quality Check

Key fixes included standardizing agency and vendor names, cleaning ZIP code values by fixing formatting issues and invalid entries, removing symbols and commas from payment amounts, converting columns into the correct data types, and replacing blank values with nulls.

DATA ENRICHMENT

Python Data Enrichment

ographic context and engineered fields to support
ryland spending



Zip Code Merging

I merged the spending data with a ZIP code reference dataset by matching vendor ZIP code to ZIP code in the location dataset. This added city, state, county, latitude, and longitude fields for each record.

Engineered Features

In_vs_out
Classified vendors as “In State” or “Out of State”

Neighbor_status
Groups vendors as Maryland, a Maryland border state, or non bordering state

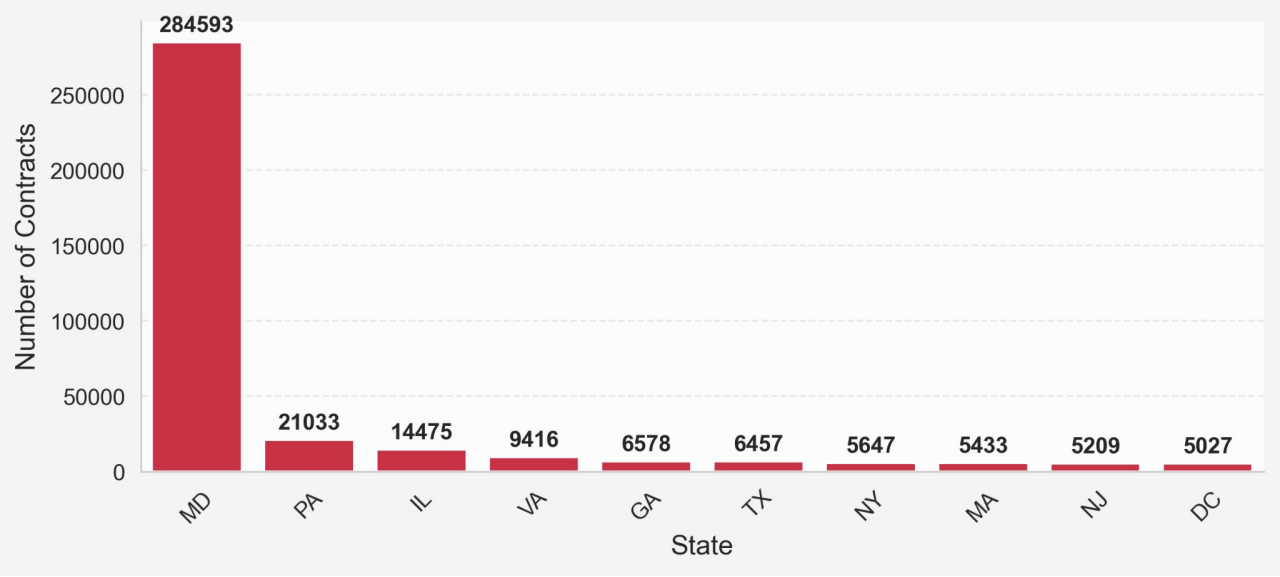
Amount_bin
Groups payments into size ranges for easy comparison (Below 1k, 1k-10k, 10k-100k, 100k+)

Why it mattered

This enrichment made it possible to analyze where Maryland payments were going, compare local vs. non-local spending, and support clearer geographic visualizations in Tableau.

EXPLORATORY ANALYSIS

Top States by Number of Vendor Contracts

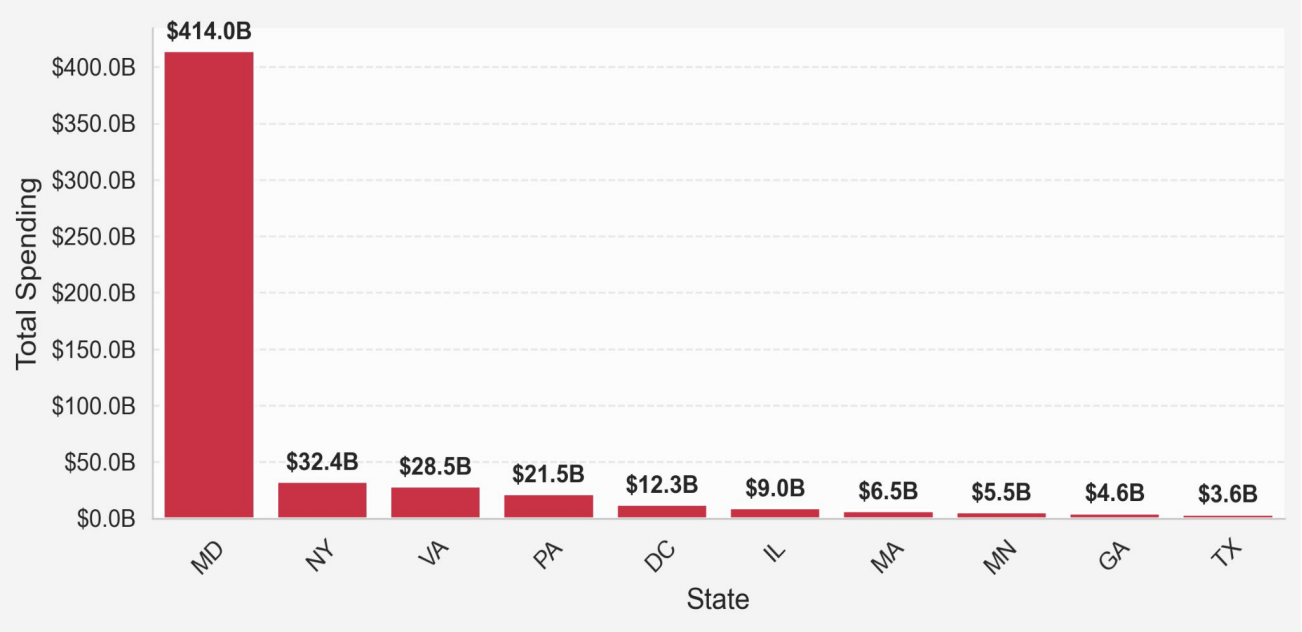


Maryland had by far the most vendor payments, showing that most payment activity stayed within the state.

Pennsylvania, Illinois, Virginia, and other states had much smaller counts in comparison.

EXPLORATORY ANALYSIS

Top States by Total Spending

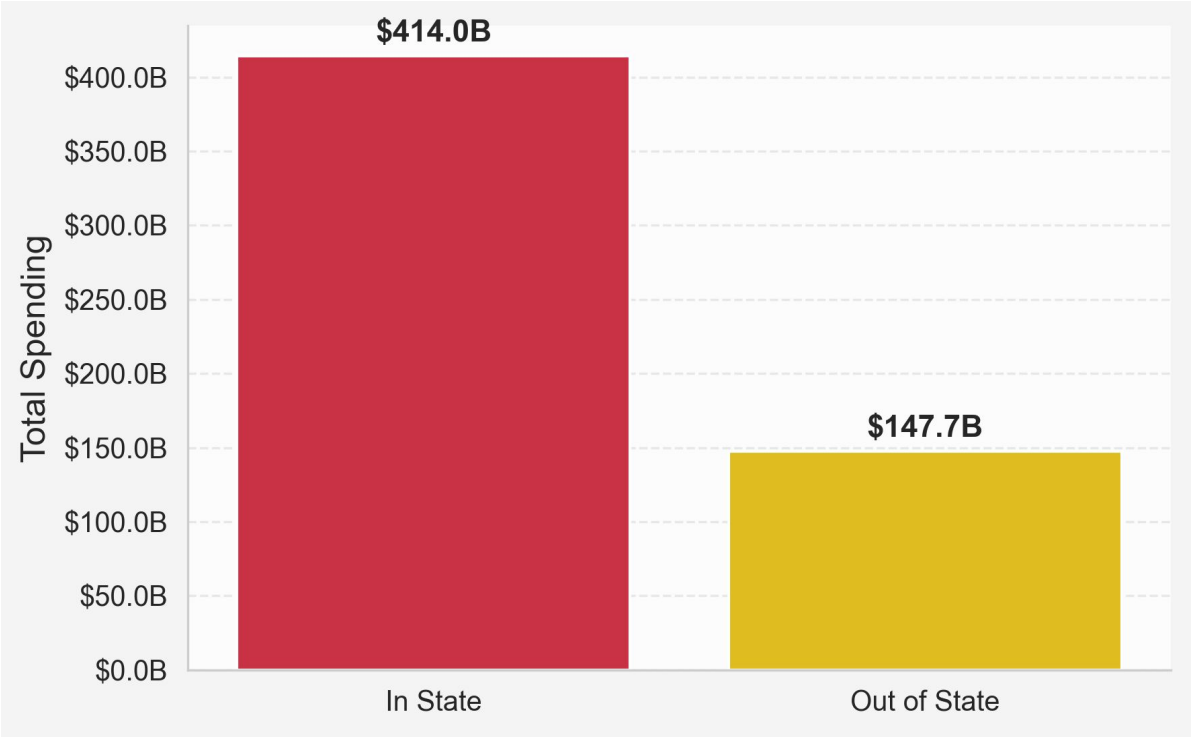


Maryland received the largest share of total spending at about $414B.

Out-of-state spending was much lower, with New York, Virginia, and Pennsylvania receiving the highest amounts outside Maryland.

EXPLORATORY ANALYSIS

In State vs Out of State Spending

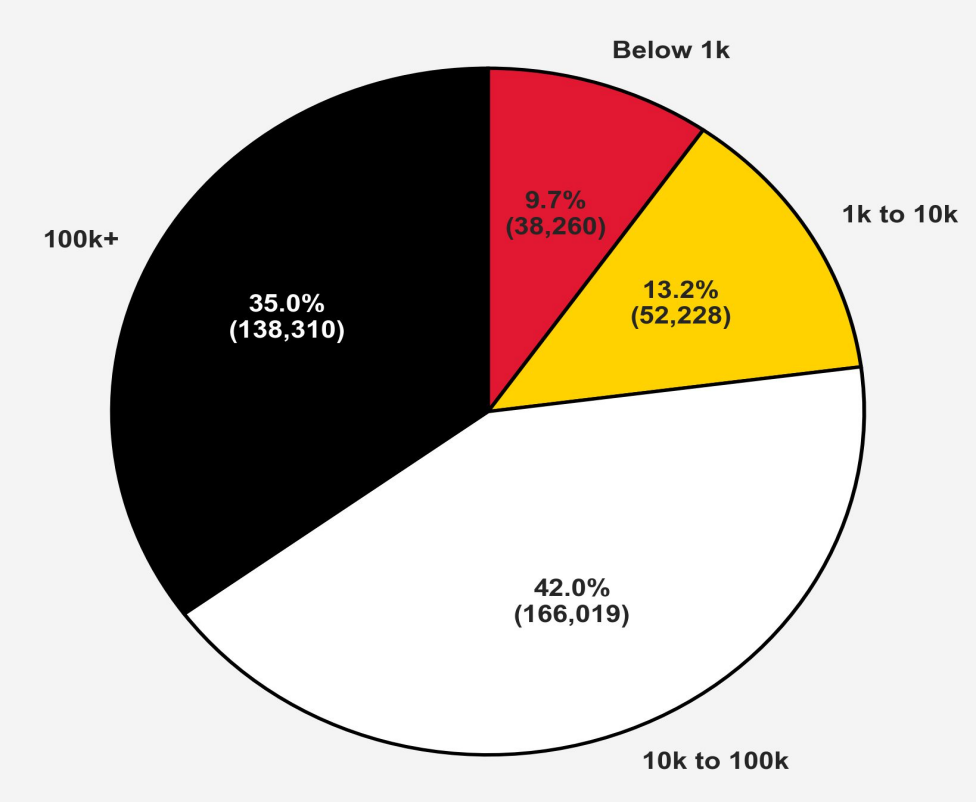


Most spending went to in-state vendors, with $414B spent in Maryland compared to $147.7B spent out of state.

This shows that a majority of Maryland's payments stayed local.

EXPLORATORY ANALYSIS

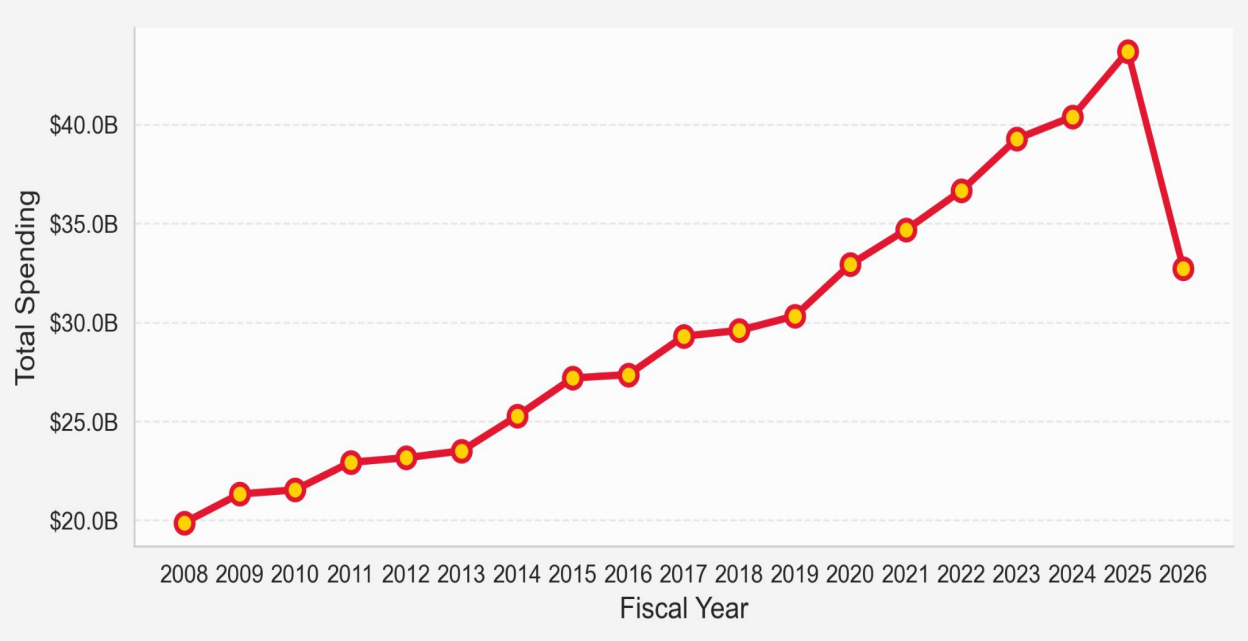
Distribution of Contract Sizes



Most payments fell in the $10K–$100K range, while payments over $100K made up a large portion as well. Smaller payments under $10K were less common overall.

EXPLORATORY ANALYSIS

Total Spending Over Time



Total spending generally increased from 2008 to 2025, with spending going up by an average of 3.12% per year. Spending reached its highest point in 2025 before dropping in 2026, which is most likely because the 2026 fiscal year data is incomplete.

Recommendations

Build a public-facing spending dashboard

Maryland should create and publish an interactive spending dashboard on its official government website to improve transparency. A dashboard like the one developed for this project would let users explore spending by fiscal year, agency, vendor, location, and payment size without needing to download and clean the raw dataset themselves. This would make the data easier for the public to understand and use.

Improve Category Code Documentation

The “category” field appears to use internal spending codes, which limits how useful the field is for public analysis. Maryland should either include a clear category reference table or plain-language category descriptions so users can better understand what each payment relates to.

Improve ZIP Code Data Entry

Data overseers should place a higher priority on ZIP code validation during data entry. Missing, invalid, or shortened ZIP codes made location matching harder, so adding stronger checks for consistent five-digit ZIP values would make it easier to connect payments to city, state, county, and map data.